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NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

First Semester B.Tech (CBCS) Degree Examinations

December 2023

PH1004-1 – QUANTUM COMPUTING AND MODERN PHYSICS

(For AD, AM, CB, CC, CS, IS, RI)

uration: 3 Hours | Max. Marks: 100

ote:

art – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each uestion carries equal marks.

art – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **nit – I & Unit – II each** and **One full** question from **Unit – III.**

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

What is a vector space?

A) A space consisting of only vectors
B) A set of vectors closed under addition and scalar multiplication
C) A space that includes both vectors and scalars
D) A space that is always three-dimensional

In a finite-dimensional vector space, the number of linearly independent vectors a basis can have is

A) 0
B) 1
C) The dimension of the vector space
D) The size of the vector space

Moore's Law originally stated that the number of transistors on a microchip would double approximately every:

A) 6 months
B) 1 year
C) 2 years
D) 3 years

What is the fundamental unit of information in quantum computing?

A) Bit
B) Byte
C) Qubit
D) Quantum gate

Which property allows qubits to represent multiple states simultaneously in quantum computing?

A) Superposition
B) Entanglement
C) Interference
D) Tunneling

The eigenvalue of a Hermitian matrix is

A) Complex conjugate
B) Real
C) Both real and complex
D) Complex

Quantum Not Gate is equivalent to

A) Pauli-X gate
B) Pauli-Y gate
C) Pauli-Z gate
D) Phase gate

Which is not a quantum gate

A) S-gate
B) T-gate
C) Hadamard gate
D) NAND gate

Which of the following is an example of an intrinsic semiconductor?

A) Silicon with added boron impurities
B) Silicon with added phosphorus impurities
C) Pure, undoped silicon
D) Silicon with added arsenic impurities

). What is the primary charge carrier in an N-type semiconductor?

A) Protons
B) Electrons
C) Holes
D) Neutrons

. When a P-type semiconductor and an N-type semiconductor are joined together to form a PN junction, what happens to the majority carriers near the junction?

A) They all disappear.
B) They combine and recombine.
C) They diffuse across the junction.
D) They become immobile.

12. What is the primary function of a solar cell?
A) To store electrical energy B) To convert sunlight into electricity
C) To generate heat from sunlight D) To provide shade and reduce sunlight

13. In the Fermi-Dirac distribution function, as temperature increases:
A) The distribution becomes narrower B) The distribution becomes broader
C) The distribution remains unchanged D) The distribution becomes linear

14. The Meissner effect in superconductors refers to the phenomenon where a superconductor:
A) Generates a strong magnetic field B) Expels a magnetic field from its interior
C) Allows magnetic field lines to pass through it easily D) Becomes a permanent magnet

15. Superconductivity is a phenomenon where certain materials exhibit:
A) Perfect electrical conductivity at all temperatures B) Perfect electrical conductivity at extremely low temperatures
C) Enhanced electrical resistance at low temperatures D) Zero electrical resistance at room temperature

16. Type I superconductors exhibit a sudden transition to the superconducting state at the critical temperature and:
A) Do not allow magnetic fields to penetrate B) Allow partial penetration of magnetic fields
C) Completely retain their normal state properties D) Exhibit a gradual transition to superconductivity

17. A laser beam is characterized by its coherence. Coherence refers to:
A) The intense brightness of the laser beam B) The ability of the laser beam to penetrate materials
C) The uniformity of the laser beam's color D) The phase relationship between the light waves in the beam

18. Laser active material in Nd: YAG laser is
A) Neodymium B) Yttrium
C) Aluminum D) Garnet

19. Which type of optical fiber has a core with a refractive index that gradually decreases from the center outward?
A) Multi-mode fiber B) Graded-index fiber
C) Single-mode fiber D) Step-index fiber

20. Which wavelength range of light is commonly used for telecommunications over optical fibers?
A) Ultraviolet B) Visible
C) Infrared D) X-ray

PART - B: DESCRIPTIVE ANSWER QUESTIONS

		Marks	BT*	CO*	PO
	Unit – I				
1. a)	Write any four differences between classical and quantum computing.	4	L2	1	1,2
b)	Explain Pauli matrices and the interaction of Pauli matrices on $\|0\rangle$ and $\|1\rangle$ state.	8	L2	1	1,2
c)	$\|\psi\rangle = A(2\|0\rangle + 3i\|1\rangle)$ find $\langle\psi\|\psi\rangle$.	4	L3	1	1,2
2. a)	What is Linear vector space? Explain its axioms.	6	L1	1	1,2
b)	Explain S gate. How does it operate on qubits? Write the truth table for it.	6	L2	1	1,2
c)	Check if the following matrix is unitary or not $A = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 & 1+i \\ 1-i & 1 \end{bmatrix}$	4	L3	1	1,2
3. a)	Explain Quantum superposition, Orthogonality, and Orthonormality of quantum states	6	L2	1	1
b)	Write a note on Bloch Sphere with a diagram.	6	L2	1	1
c)	Verify, if $A=\begin{bmatrix} 1 & 4+3i \\ 4-3i & 5 \end{bmatrix}$ and $B=\begin{bmatrix} 1 & 4-3i \\ 4-3i & 5 \end{bmatrix}$ are Hermitian or not.	4	L3	1	

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Unit – II

4.	a)	Explain the probability of occupation for the energy levels $E<E_F$ and $E>E_F$ at T=0K,for metals.	6	L2	2	1,2
	b)	From carrier concentration, deduce the expression for electrical conductivity of an extrinsic semiconductor.	6	L2	2	1,2
	c)	Calculate the resistivity of intrinsic germanium if the intrinsic carrier density is 2.5×10^{19} m^{-3} assuming electron and hole mobilities of 0.38 $m^2v^{-1}s^{-1}$ and 0.18 $m^2v^{-1}s^{-1}$ respectively.	4	L3	2	1,2
5.	a)	Write a detailed note on solar cells.	4	L2	2	1,2
	b)	What is the Hall effect? Explain the generation of the Hall field in the semiconductor. Obtain expressions for carrier concentration and Hall coefficient.	8	L2	2	1,2
	c)	Calculate the probability of occupation of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K.	4	L3	2	1,2
6.	a)	What is superconductivity? Explain the important characteristic properties of superconductors.	6	L2	3	1,2
	b)	Explain the magnetic behaviour of Type-I and Type-II superconductors.	6	L2	3	1,2
	c)	The critical temperature and critical magnetic field for superconducting lead are 7.2 K and 800 gauss respectively. What will be the temperature up to which lead will be in superconducting state in a magnetic field of 400 gauss?	4	L3	3	1,2

Unit – III

7.	a)	Differentiate between spontaneous and stimulated emission. Which of them is important in laser action and why?	4	L2	4	1,2
	b)	Explain the construction and working of a Nd-YAG laser with neat diagrams.	8	L2	4	1,2
	c)	Find the ratio of population of two energy states, the transition between which is responsible for the emission of photons of wavelength 694.3nm at temperature 300 K.	4	L4	4	1,2
8.	a)	What is an optical fiber? Derive an expression for the numerical aperture of an optical fiber.	6	L3	5	1,2
	b)	Explain the different types of multi-mode optical fibers with suitable diagrams.	6	L2	5	1,2
	c)	Calculate the numerical aperture and acceptance angle of an optical fiber with core and cladding refractive indices of 1.45 and 1.40 respectively.	4	L4	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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Duration: 3 Hours | Max. Marks: 100

List of constants: Velocity of light, $c=3\times10^{8}ms^{-1}$, Planck's constant, $h=6.63\times10^{-34}$ Js, Electron mass, $m=9.11\times10^{-31}$kg, Electron charge, $e=1.602\times10^{-19}$C, Permittivity of vacuum, $\varepsilon_o = 8.85\times10^{-12}$ F/m, Boltzmann constant, $k=1.38\times10^{-23}$J/K. Avogadro number, $N_A = 6.023 \times 10^{26}$/ kg mole.

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II each** and **One full** question from **Unit – III.**

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **A qubit is used represent a _____quantum-mechanical system.**
   A) One-state B) Two-state
   C) Three-state D) Four-state

2. **The set of vectors which follow the same properties followed by linear vector space is said be**
   A) Basis B) Dimension
   C) Hilbert space D) Orthogonal state

3. **It is the process of replacing i^{th} row of the matrix by i^{th} column, then it is said to be**
   A) Conjugate Matrix B) Transpose Matrix
   C) Identity Matrix D) Hermitian Operator

4. **Which property allows qubits to represent multiple states simultaneously in quantum computing?**
   A) Superposition B) Entanglement
   C) Interference D) Tunnelling

5. **$|0\rangle$ and $|1\rangle$ are orthogonal if:**
   A) They are perpendicular B) They are parallel
   C) Angle between them is 0 D) Linearly independent

6. **In $|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$, α, β represents,**
   A) Ground state and excited state B) Probability density
   C) Probability amplitude D) Normalization

7. **What does 'entanglement' mean?**
   A) Two particles are different B) Two particles are separate
   C) Two particles are independent D) Two particles are connected

8. **In a quantum circuit, this gate is used to place a qubit into superposition.**
   A) Hadamard B) X-gate
   C) Bloch D) CNOT

9. **An elemental semiconductor has generally ..... valence electrons.**
   A) 2 B) 3
   C) 6 D) 4

10. As the doping to a pure semiconductor increases, the bulk resistance of the semiconductor .....
A) Remains the same B) Increases
C) Decreases D) Becomes zero

11. In a N-type semiconductor, the position of Fermi-level is
A) Close to the valance band B) In the middle of the energy band gap
C) Can be any where D) Close to the conduction band

12. Measurement of Hall coefficient enables the determination of:
A) Temperature coefficient and thermal conductivity B) Mobility and carrier concentration
C) Fermi level and forbidden energy gap D) Area of the device

13. The density of charge carriers in a pure semiconductor is proportional to
A) exp(-Eg/kT) B) exp(-2Eg/kT)
C) exp ((-Eg/kT²) D) exp(-Eg/2kT)

14. The primary function of a solar cell is
A) To store solar energy in chemical form B) To reflect sunlight to solar heaters
C) To convert sunlight directly into electrical energy D) To absorb and retain heat from sunlight

15. When there is no incident light, the reverse current in a photodiode is essentially non-existent and is referred to as
A) Zener current B) Dark current
C) Photocurrent D) PIN current

16. The magnetic susceptibility (χ) in superconductor is
A) Positive B) Zero
C) Negative D) Infinity

17. Emission of a photon by an excited atom due to interaction with a passing photon nearby is called
A) Stimulated emission B) Induced absorption
C) Spontaneous emission D) Thermionic emission

18. The purpose of the optical cavity in a laser is to:
A) Convert electrical energy into light B) Amplify the light through multiple passes between mirrors
C) Filter out unwanted frequencies of light D) Focus the light into a beam

19. The numerical aperture (NA) in optical fiber is used to describe
A) Light gathering or light-collecting ability B) Light spreading ability
C) Light output from an external shield D) Light leakage ability

20. The loss in signal power as light travels down a fiber is called
A) Dispersion B) Scattering
C) Absorption D) Attenuation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I	Marks	BT*	CO*	PO*
1. a) Write any four differences between classical and quantum computing.	4	L*2	1	1, 2
b) Explain the interaction of Pauli X, Y, Z matrices on $\|0\rangle$ and $\|1\rangle$ states.	8	L2	1	1, 2
c) If $1\alpha> = \binom{a}{b}$ and $1\beta> = \binom{c}{d}$ then prove that $\langle\alpha\|\beta\rangle = \langle\beta\|\alpha\rangle^*$	4	L3	1	1, 2
2. a) What is a qubit? Write any four properties of qubits.	4	L2	2	1, 2
b) Explain the operation on quantum states for the following gates with the truth table: (i) Hadamard Gate (ii) S gate	8	L2	2	1, 2
c) $\|\psi\rangle = A\,(2\|0\rangle + 3i\|1\rangle)$ find $\langle\psi\|\psi\rangle$	4	L3	2	1, 2

3.	a)	Write the fundamental postulates of quantum mechanics.	4	L2	2	1, 2
	b)	What is a Bloch sphere? Explain the representation of qubits on Bloch sphere.	8	L2	2	1, 2
	c)	Verify if $\|w\rangle = \begin{pmatrix}1\\1\end{pmatrix}$ & $\|v\rangle = \begin{pmatrix}1\\-1\end{pmatrix}$ are orthogonal.	4	L3	2	1, 2

Unit – II

4.	a)	What is an intrinsic semiconductor? Derive an expression for the conductivity of an intrinsic semiconductor.	6	L2	3	1, 2
	b)	Explain the effect of temperature on the Fermi level in an extrinsic semiconductor.	6	L2	3	1, 2
	c)	Mobilities of electrons and holes in a sample of intrinsic germanium at 300 K are 0.36 $m^2 V^{-1}S^{-1}$ and 0.17 $m^2 V^{-1}S^{-1}$ respectively. If the resistivity of the specimen is 2.12 Ωm, compute the intrinsic carrier density.	4	L3	3	1, 2
5.	a)	Explain the construction and working of solar cell.	4	L2	3	1, 2
	b)	State Hall Effect. Explain how Hall field is set up in the given semiconductor sample and obtain an expression for the Hall voltage and Hall coefficient of an n-type semiconductor.	8	L2	3	1, 2
	c)	The Hall coefficient of a specimen of a doped silicon is found to be 3.66×10^{-4} m^3C^{-1} and its resistivity is found to be 8.93×10^{-3} Ωm. Calculate charge carrier concentration and electron mobility at room temperature.	4	L3	3	1, 2
6.	a)	What are superconductors? Explain Meissner effect in superconductors.	4	L2	4	1, 2
	b)	Explain Type-I and Type-II superconductors with suitable diagrams.	8	L2	4	1, 2
	c)	The critical temperature and critical magnetic field at 0K for superconducting lead are 7.2 K and 800 gauss respectively. What will be the temperature up to which lead will be in superconducting state in a magnetic field of 400 gauss?	4	L3	4	1, 2

Unit – III

7.	a)	What is a laser? Explain the principle of stimulated emission of radiation.	4	L2	5	1, 2
	b)	With neat diagrams, describe the construction and working of Nd:YAG laser.	8	L2	5	1, 2
	c)	Calculate the wavelength of emission from a GaAs semiconductor laser whose band gap energy is 1.44 eV.	4	L3	5	1, 2
8.	a)	What is an optical fiber? Describe how the optical fiber can be used in communication system.	4	L2	5	1, 2
	b)	With diagram derive an expression for numerical aperture of an optical fiber in terms of R.I. of core and cladding.	8	L2	5	1, 2
	c)	A fiber has a core diameter of 6 μm and its core refractive index is 1.47 and for cladding it is 1.43. How many modes can propagate into the fiber if the wavelength of the source is 1.5 μm.	4	L3	5	1, 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome

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